Franklin Muoghalu

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Education

University of Houston — May 2026

Bachelor's in Computer Science

- GPA of **3.57/4.0**; Dean's list student

Technical Skills

Programming Languages: Python, C++, C, C#, JavaScript, HTML/CSS
Frameworks & Libraries: React.js, Node.js, Express.js, Flask, FastAPI, ASP.NET
Tools & Platforms: Git, GitHub, GitHub Actions, Docker, Google Cloud, Mysql, Postgres
Server & DevOps: Nginx, Reverse Proxy Configuration, SSL/TLS, Cloudflare DNS, SSH, Systemd Services

Work Experience

Software Engineer, Foxconn — May 2026– Present

- Architected an asynchronous PostgreSQL database configuration listener utilizing Python's arq worker and database triggers, enabling real-time background updates to system cron schedules
- Designed and deployed multi-tier architectures and microservices by embedding independent Nginx server configurations directly inside individual Docker container layers, ensuring strict network isolation and secure public routing
- Established live-site telemetry and logging protocols, writing Troubleshooting Guides (TSGs) to accelerate incident resolution and reduce system downtime.
- Developed and maintained multiple backend routing configurations and database schemas across different teams using FastAPI, Flask, and ASP.NET Core

System Engineer, Foxconn — May 2025 – May 2026

- Designed and deployed internal testing software using Python and C#, automating manual processes to increase overall testing efficiency by **30%**
- Engineered a suite of **5** custom full-stack applications with React to resolve critical operational bottlenecks across different departments
- Authored comprehensive technical documentation and system guidelines, accelerating technician workflows and reducing new-hire onboarding time by **20%**

Project Experience

Self-Hosted Server Infrastructure & Deployment — October 2024 – Present

- Technologies Used: **Nginx, Docker, Cloudflare, Cloudflare Tunnels, Linux, Oracle Cloud, Wireguard**
- Built and maintained a multi-platform server infrastructure spanning a Raspberry Pi 5, a home Linux server, and an Oracle Cloud instance
- Hosted and managed **8** applications including personal web apps, LLM tools, and a multiplayer game server, ensuring high availability and secure access
- Configured a full Nginx-based reverse proxy with Cloudflare SSL/TLS, DNS, and firewall protections across all environments
- Deployed database services on Oracle Cloud, supporting API backends and real-time application features through optimized connection management
- Implemented automated updates, containerized services, and monitoring tools to streamline maintenance and system reliability

Bone Pile Management System & Documentation System — March 2026 – May 2026

- Technologies Used: **Python, FastAPI, Express.js, React, Docker, PostgreSQL, LLM, RAG**
- Designed and built an AI-driven management platform that automatically logs, categorizes, and tracks hardware errors and board data within the bone pile
- Implemented a hybrid backend architecture utilizing FastAPI and Express.js to handle efficient CRUD operations, database routing, and file processing
- Integrated a locally running RAG model with LLM and NLP components to provide engineers with real-time contextual suggestions, error explanations, and automated documentation generation
- Boosted overall efficiency of the failure analysis department by 25% through centralized issue tracking and AI-assisted debugging workflows

O.M.N.I — November 2024 - May 2025

- Technologies Used: **Python, LLMs, Embeddings, NLP, RAG, OpenAI, Docker, Cloudflare**
- Won the **Grand Prize** at the Johnson Space Center Hackathon for applying O.M.N.I to NASA mission manuals, enabling astronauts to rapidly retrieve emergency procedures using natural-language queries
- Built an intelligent AI agent with a Python backend and Tkinter UI, designed to allow users to query and navigate complex technical information in real time
- Developed a publicly accessible API capable of ingesting documents from any source, generating embeddings, and storing them in a vector database for high-accuracy retrieval
- Containerized and deployed the system using Docker and Cloudflare Tunnels for secure, remote accessibility and testing